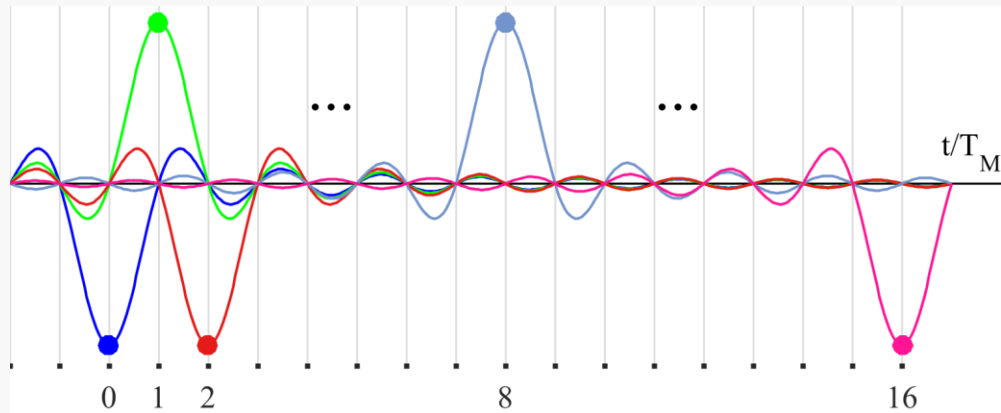


Linear Modulations

Qasim Chaudhari



Digital Electronics

Concept behind simple digital logic

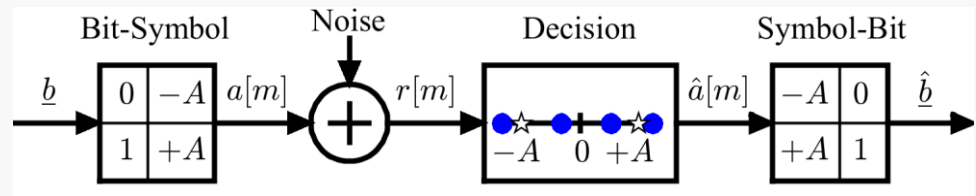
- Logic 0 can be assigned to one voltage level while logic 1 to another
- System electronics only need to differentiate between these two levels

Digital information is a sequence of 0's and 1's

A Simple Communication System

Symbols: 0's and 1's are assigned to two distinct voltage levels $-A$ and $+A$

$$a[m] = \begin{cases} -A, & b = 0 \\ +A, & b = 1 \end{cases}$$



Rx performs the inverse mapping from voltage to bits

Minimum Distance Rule

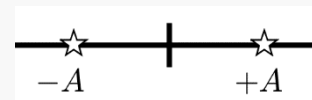
... we balance probabilities and choose the most likely.

Sherlock Holmes - The Hound of the Baskervilles

Decision on Rx symbol is made based on whether it is closer to $-A$ or $+A$

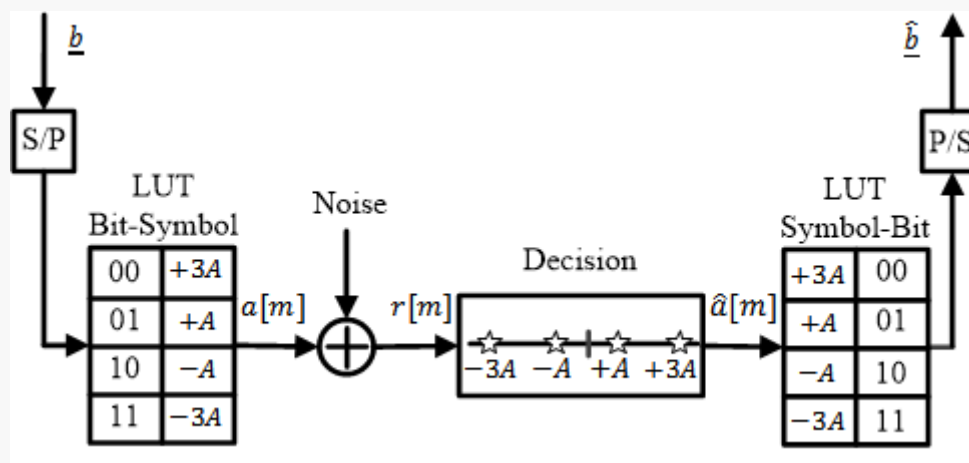
Minimum distance rule

$$\hat{a}[m] = \begin{cases} -A, & r[m] < 0 \\ +A, & r[m] > 0 \end{cases}$$



Packing More Bits in One Symbol

- A symbol is a building block of a digital communication system
- If our Rx can differentiate between two signal levels, it can certainly differentiate between more than two signal levels as well
- Bit-symbol mapping



Some Real World Channels

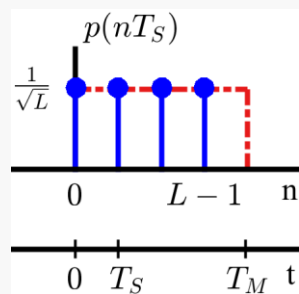
- wireless
- telephone lines
- coaxial cable
- optical fiber
- ethernet
- USB
- chips on a printed circuit board



Symbol sequence is just a set of numbers and cannot be sent on a real channel

From Numbers to Signals

- An analog waveform is required!
- Start with a rectangular pulse shape (NRZ) $p(nT_S)$



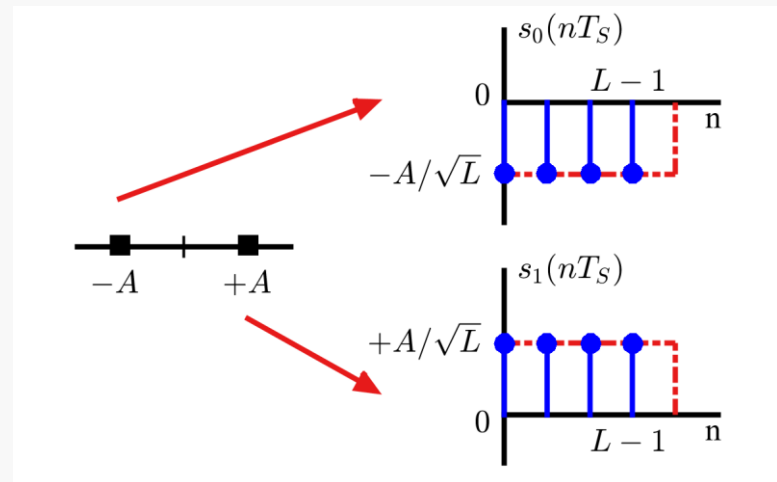
Pulse duration is equal to one **symbol interval** $T_M = L \cdot T_S$

L is the number of samples/symbol

Amplitude Modulation

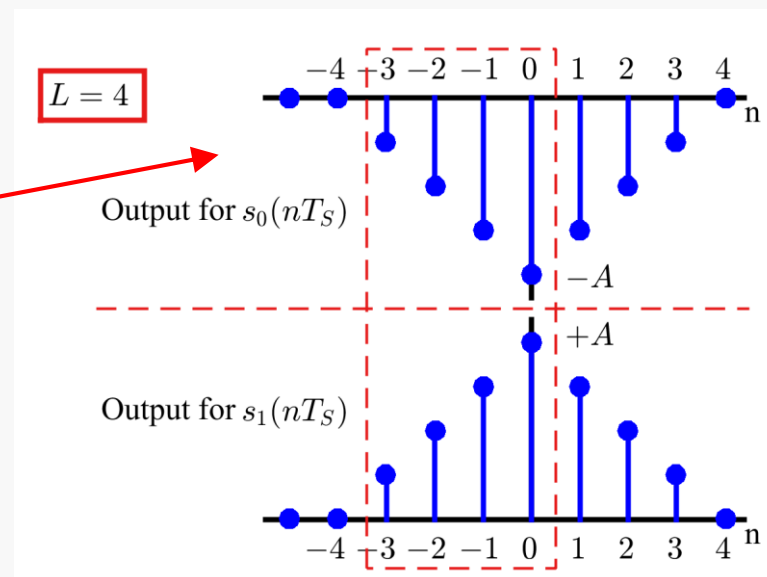
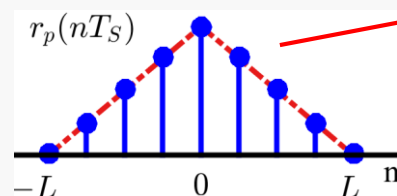
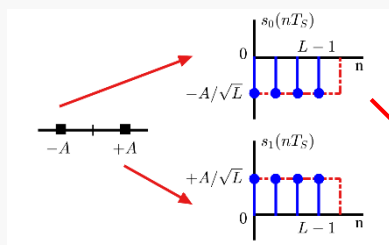
Modulation: Changing some characteristic of $p(nT_S)$ as a function of bits

Here, we are changing the amplitudes



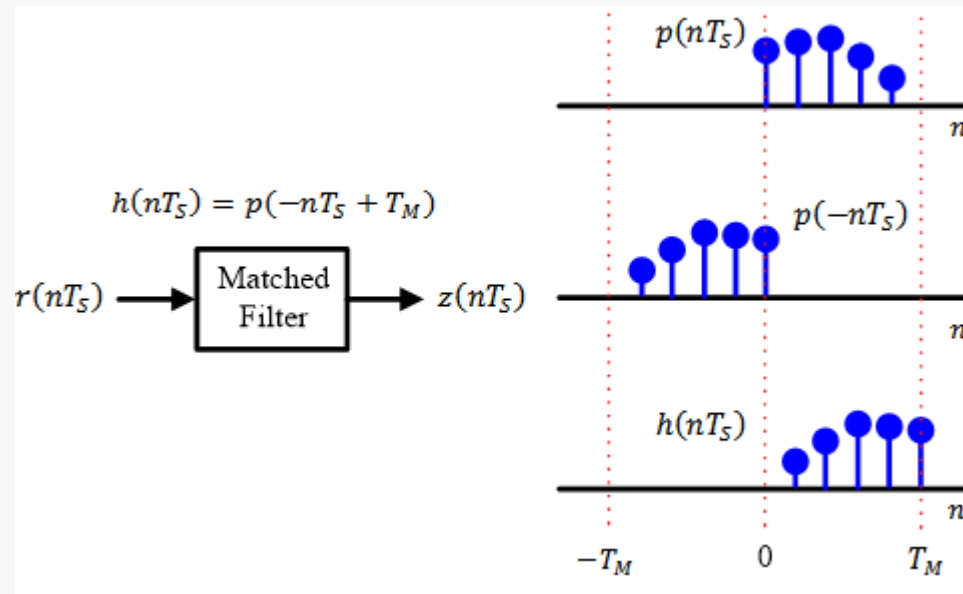
From Signals Back to Numbers

- Correlate the incoming signal $r(nT_S)$ with possible choices $s_0(nT_S)$ and $s_1(nT_S)$
- Why not exploit linearity and correlate with the pulse shape only?



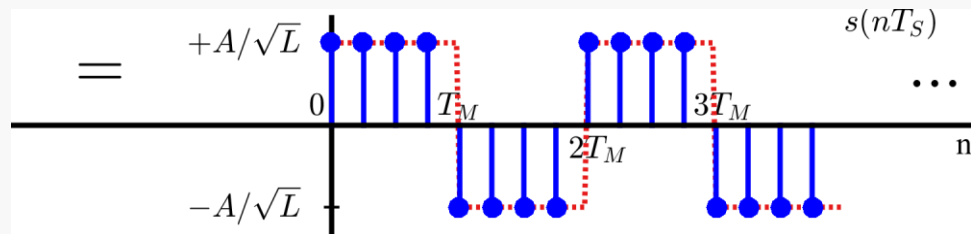
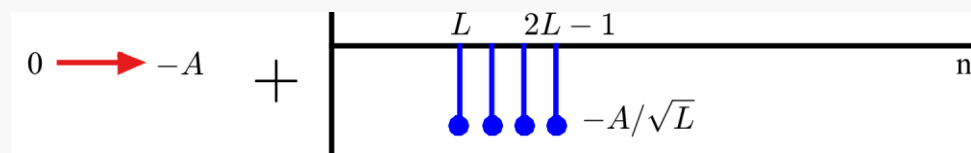
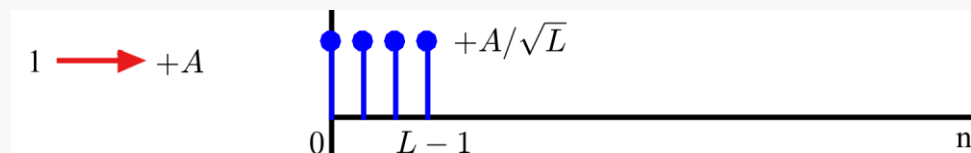
Matched Filter

Correlation of the Rx signal $r(nT_S)$ with the pulse shape $p(nT_S)$ can be interpreted as an output from a filter



Filter impulse response is a flipped version of the pulse shape $p(nT_S)$, why?

Pulse Amplitude Modulation (PAM)



$$a[0]p(nT_S - 0T_M)$$

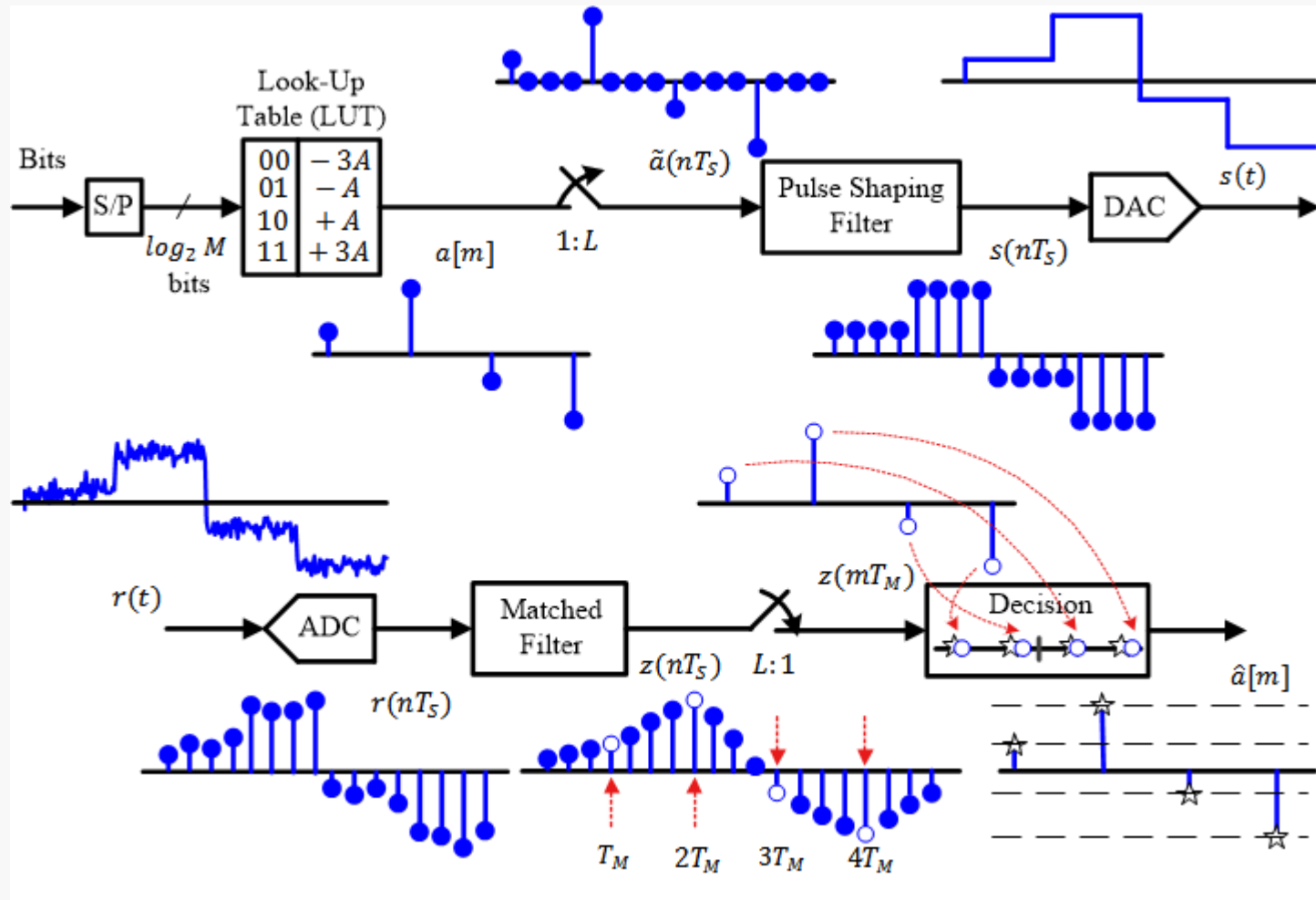
$$a[1]p(nT_S - 1T_M)$$

$$a[2]p(nT_S - 2T_M)$$

$$a[3]p(nT_S - 3T_M)$$

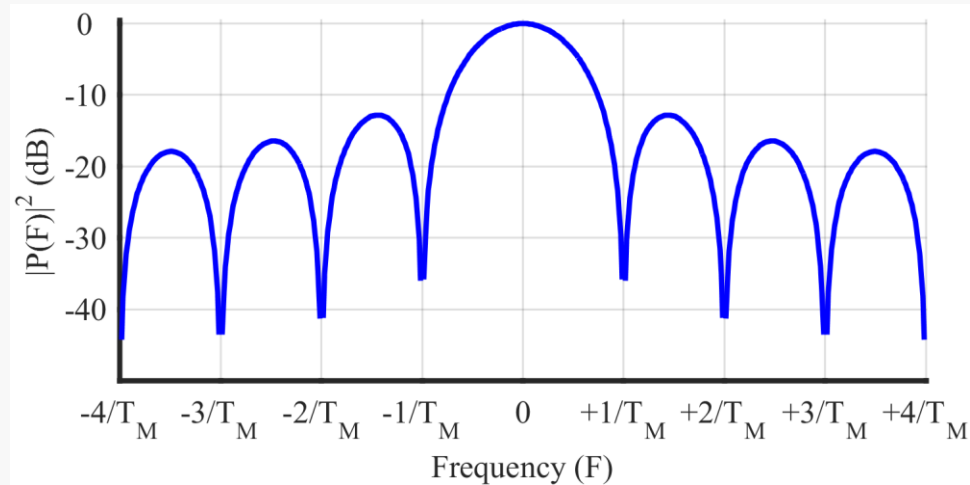
$$s(nT_S) = \sum_m a[m]p(nT_S - mT_M)$$

Pulse Amplitude Modulation (PAM)



Spectrum of a Rectangular Pulse

Wide spectrum due to high sidelobes not decaying fast enough

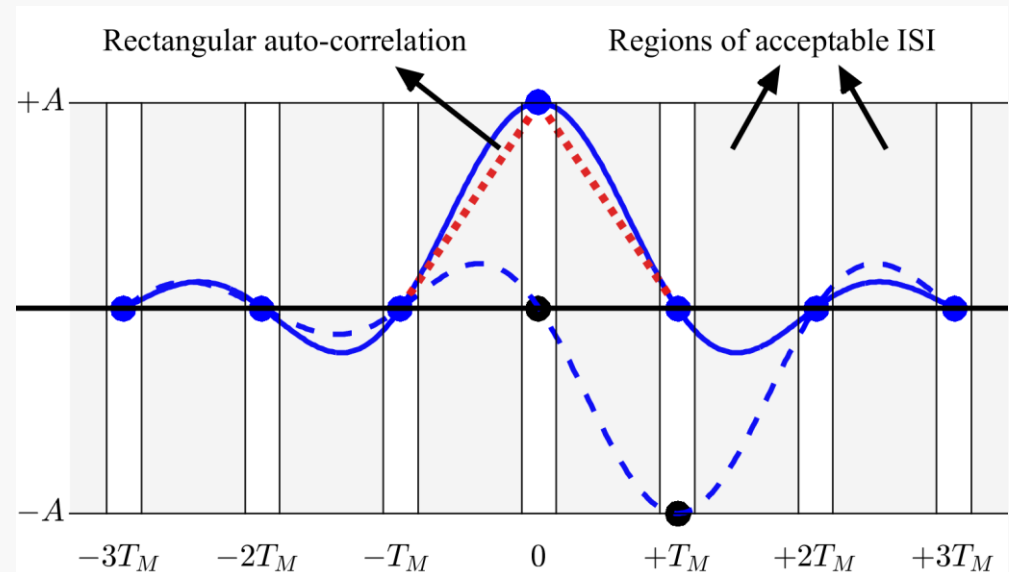


Required: a pulse extended in time

Nyquist No-ISI Criterion

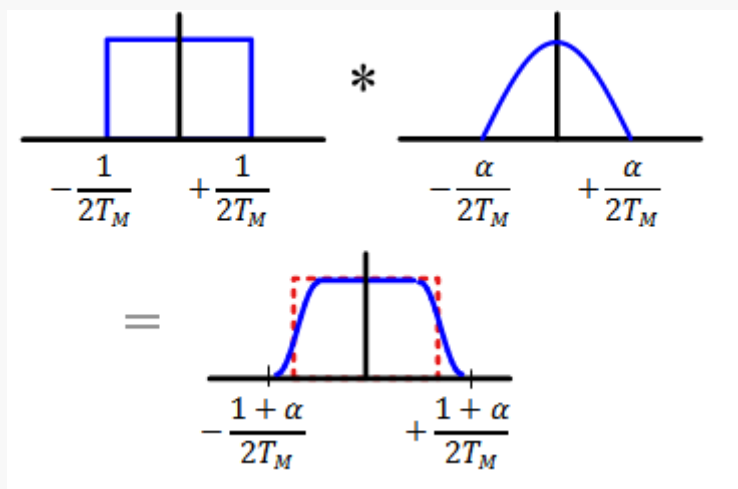
- Signals narrow in time are wide in frequency domain
- Causes interference from symbols both before and after each symbol (ISI)
- However, critical samples at the output of matched filter occur at the end of integer multiples of symbol duration T_M

$$r_p(mT_M) = \begin{cases} 1, & m = 0 \\ 0, & m \neq 0 \end{cases}$$



Increasing the Bandwidth

- Most compact spectrum is rectangular (impractical)
- Increase its bandwidth through spectral convolution
- Factor α specifies the bandwidth beyond the minimum and called **excess bandwidth** or **rolloff factor**

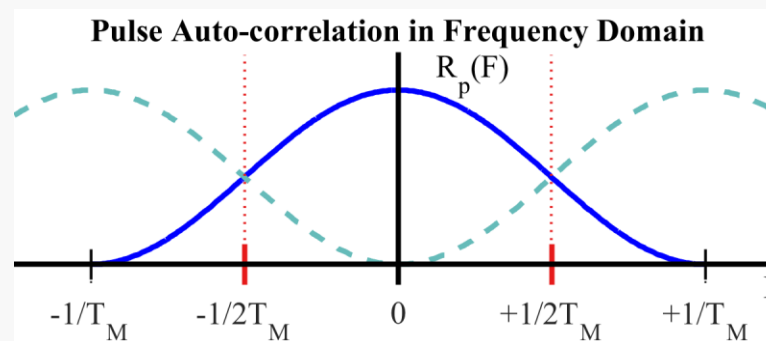


Raised Cosine Spectrum

A frequency domain cosine raised by a constant

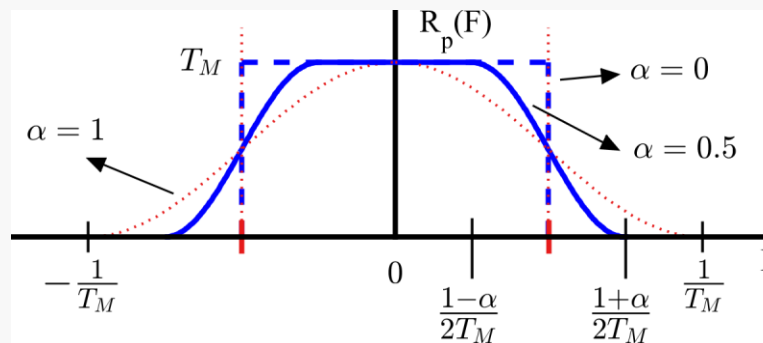
Case $\alpha = 1$

$$R_p(F) = \frac{T_M}{2} \left[1 + \cos \left(2\pi \cdot \frac{T_M}{2} \cdot |F| \right) \right] \quad -\frac{1}{T_M} \leq F \leq +\frac{1}{T_M}$$



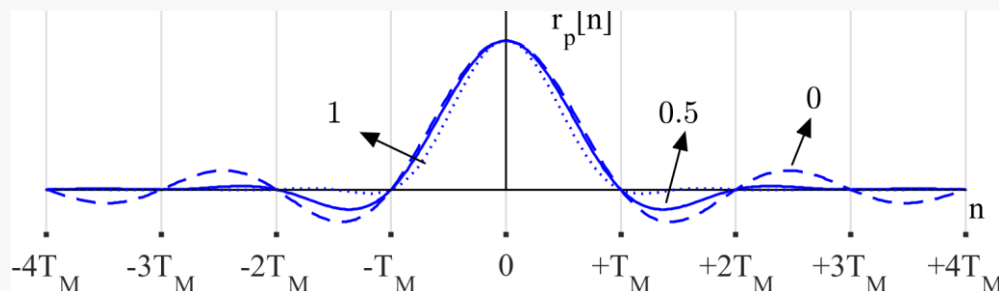
Raised Cosine Spectrum

For a general α



Impulse response

$$r_p[n] = \frac{\sin(\pi n/L)}{\pi n/L} \cdot \frac{\cos(\pi \alpha n/L)}{1 - (2\alpha n/L)^2}$$



Square-Root Raised Cosine

Where to place raised cosine?

- Rx: spectral leakage cannot be controlled
- Tx: extra interference and noise entering the Rx system
- Both: pulse can be split into two parts, one at the Tx and the other at the Rx

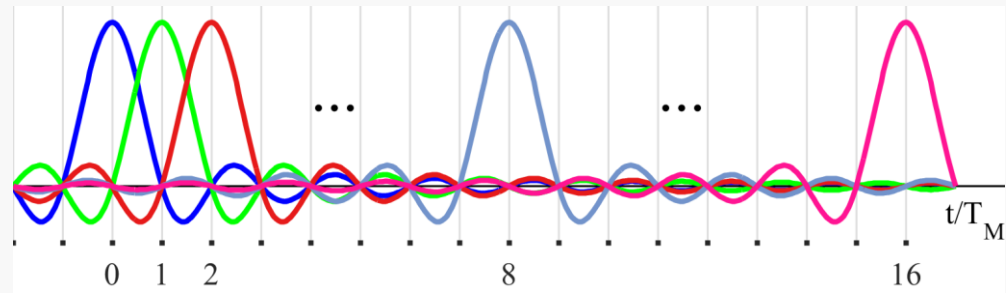
$$r_p(nT_S) = p(nT_S) * p^*(-nT_S)$$

$$R_p(F) = P(F) \cdot P^*(F) = |P(F)|^2$$

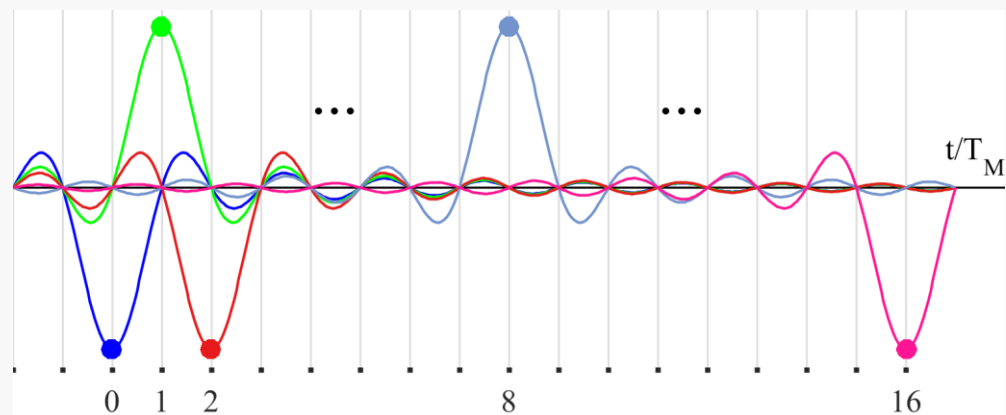
$$P(F) = \sqrt{R_p(F)}$$

Role of Pulses

Unmodulated



Modulated



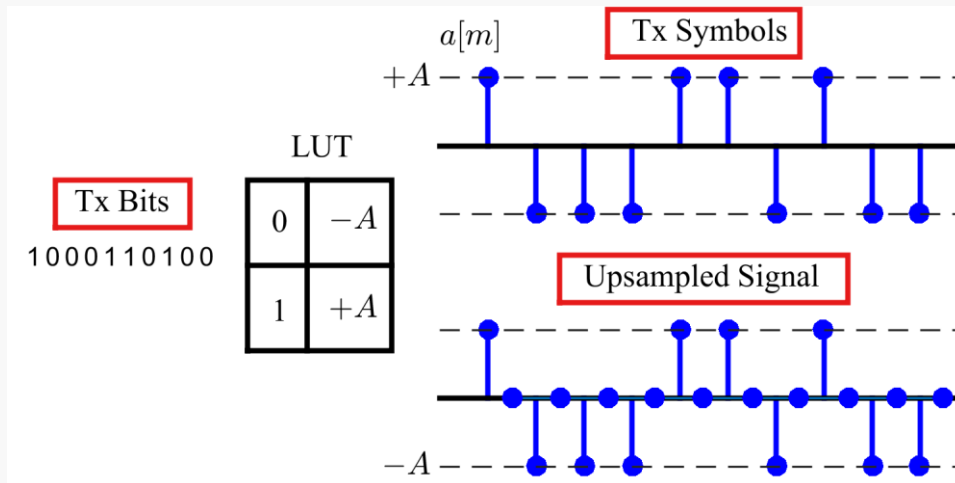
'Root Raised Cosine Filter' block in GRC

Tool firderes with member function

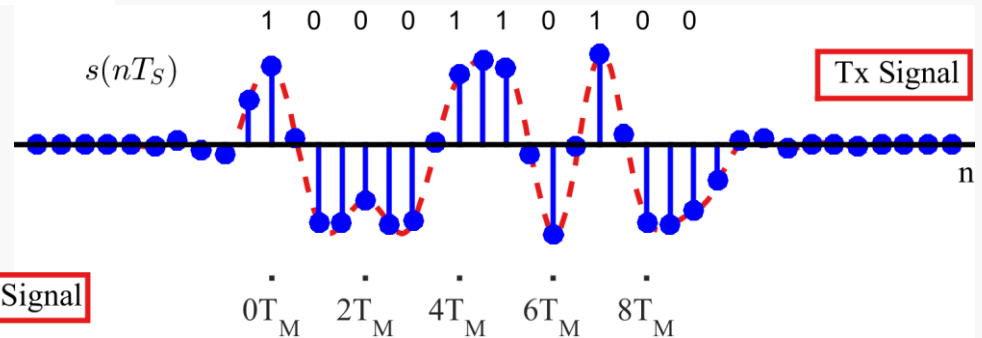
```
static std::vector<float> gr::filter::firderes::root_raised_cosine (  
    double  gain,  
    double  sampling_freq,  
    double  symbol_rate,  
    double  alpha,  
    int      n taps  )
```

PAM Revisited

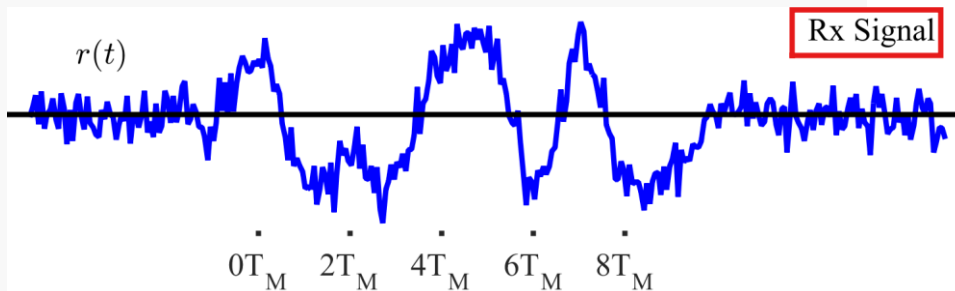
1. Signal generation



2. Pulse shaping

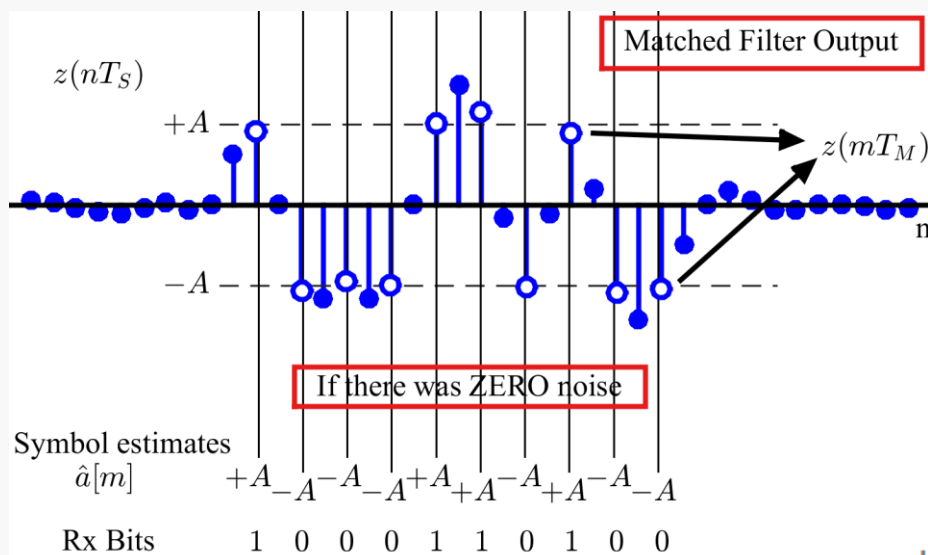


3. Rx signal

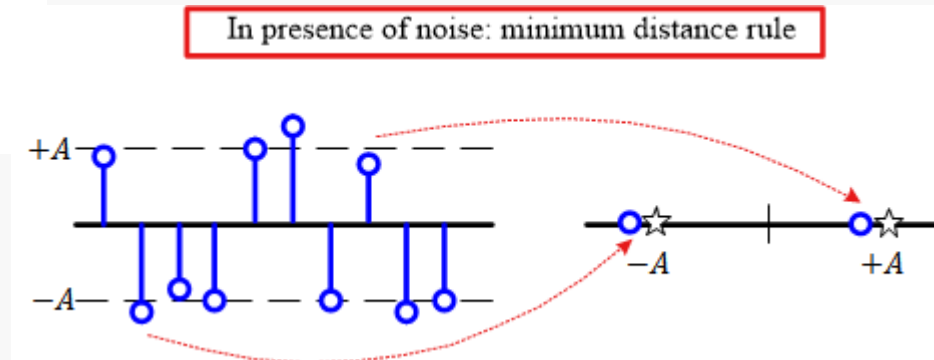


PAM Revisited

4. Matched filtering



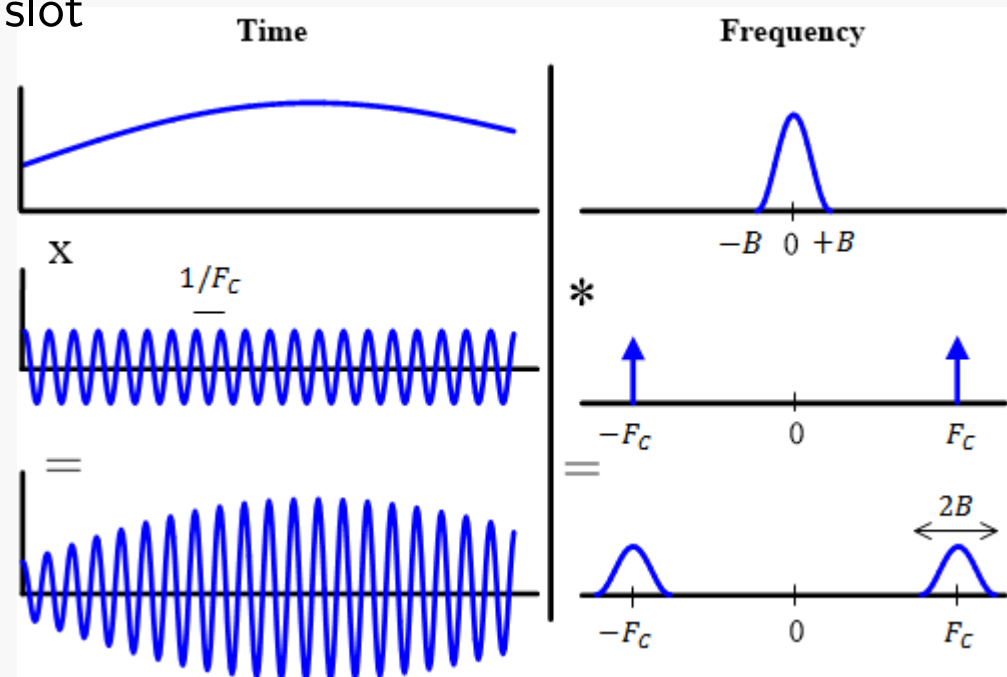
5. Detection



Take every L^{th} matched filter output and discard all the other samples

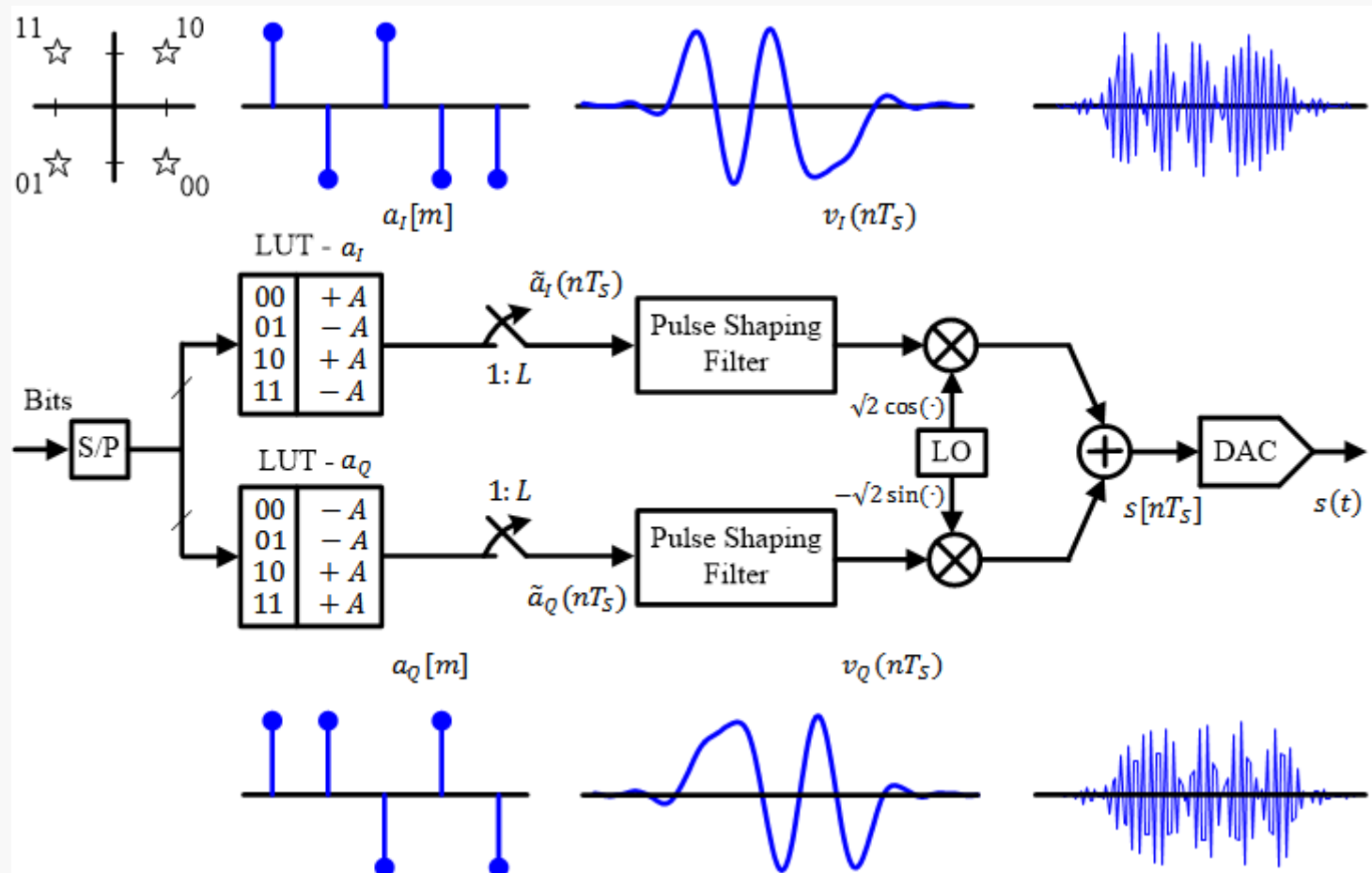
From Baseband to Passband

- Wireless transmission through electromagnetic waves
- PAM waveform can be multiplied with a carrier sinusoid at frequency F_C
- Time domain multiplication is convolution between their spectra
- Signal ``parked'' at its allocated slot



Quadrature Amplitude Modulation (QAM)

Two carriers, a cosine and a sine, carry independent waveforms (orthogonality)



Quadrature Amplitude Modulation (QAM)

